

RESEARCH ARTICLE

L-MACD: Learning-Based Adaptive CAPTCHA Defense Against Multimodal AI Attacks

MUKUL SANGWAN¹, ASTHA GUPTA,
AND B V A N S S PRABHAKAR RAO², (Senior Member, IEEE)

School of Computer Science and Engineering, Vellore Institute of Technology, Chennai, India

Corresponding author: B V A N S S Prabhakar Rao (prabhakarao@vit.ac.in)

ABSTRACT The high rate of development of multimodal AI models such as GPT-4V, Gemini 1.5, and Claude 3 has rendered the conventional CAPTCHA schemes far less efficient. In order to address this challenge, the paper introduces L-MACD (Learning-based Multi-modal Adaptive CAPTCHA Defense) a reinforcement learning architecture which modifies the difficulty of CAPTCHA in real-time. The system uses feedback as input continuously to modify its complexity to adapt to the specific user input through the simulated human user and also the AI agents to adjust its the most important parameters like visual distortion, background noise or image rotation. It predicts the most effective defense configuration to be applied in every situation using an adaptive model implemented as a random forest algorithms. Also, a new metric has been proposed, namely Adversarial Gain Equilibrium (AGE), to quantify the extent to which a system remains human-usable and resistant to automated attacks. Four configurations of the framework were tried, being Baseline (G1), LLM-ZSA-CG (G2), Human-Expert (G3) and RL-Adaptive (G4). The findings indicate that L-MACD decreases the AI success rate from 68% to 8% (approximately 85% reduction), while maintaining 85–90% human usability. A t-test supported approach to statistical analysis shows that RL-Adaptive approach is better than static base methodologies. Additional research with SHAP-based explainability shows that rotation and distortion parameters have the greatest effect on adversarial gain reduction. To summarize, L-MACD is a versatile and transparent defense mechanism, which is capable of combating the increasing threat of sophisticated multimodal AI systems.

INDEX TERMS CAPTCHA, reinforcement learning, multimodal AI, adversarial learning, explainable AI, security.

I. INTRODUCTION

The contemporary virtual world is becoming the enactment of ever-lasting struggle between robotic systems and security measures developed to detect and prevent them. In several years, the Completely Automated Public Turing Test to Make Computers and Humans Apart (CAPTCHA) has been one of the most popular and efficient measures against spamming, automated registration, etc., in the range of internet misconduct. Its underlying idea is simple yet powerful: to present a challenge that humans can solve with ease but that remains difficult for machines. Nonetheless, this principle underlining it has started to fail as the development of artificial intelligence is growing faster than ever.

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The Multimodal Large Language Models (MLLM) balance of power has changed lately due to the advancements in MLLMs. MLLMs have very good generalization to a wide variety of visual-language tasks, demonstrating high-performance at vision encoders after combining these encoders with advanced natural language reasoning. Unlike traditional CAPTCHA solvers which depend on problem-specific, carefully trained algorithms, these new models can operate as a general purpose problem solver, with only the slightest or no extra training necessary to handle an unknown CAPTCHA format.

Test experiments on the first attempts in this research present an outrageous turnaround of the intended use of CAPTCHA. On a standard CAPTCHA dataset test in text (G1\Baseline) a state-of-the-art MLLM has a success rate of 83.6%, compared to human subjects who scored 66.7%.

This finding highlights some fundamental failure in the logic of CAPTCHA, the systems that are supposed to be unique to humans and machines have been learned by machines far better. As a result, valid human users experience an increasing burden placed on them to gain access, and automated bots increasingly access information without much difficulty.

The essence of this weakness is that there are no dynamic and novel methods of generating CAPTCHA. Although such variations are caused by limited and learnable transformation patterns even in the case of adversarial methods or semantic variations are used. Analysis of the results shows that MLLMs can maintain a 83% success rate on such an adversarial CAPTCHA, which shows that static defenses can no longer defend against adaptive, general purpose AI solvers.

To address these shortcomings, this paper submits that the CAPTCHA systems should be changed to adaptive and self-learning architectures, and should be not fixed designs. We introduce the Learning-based Multi-modal Adaptive CAPTCHA Defense (L-MACD) framework and its equivalents, G4RL_ADAPTIVE, which are augmented by reinforcement learning. The suggested method uses the reinforcement based learning to automatically enforce the control of the CAPTCHA challenge by real-time feedback of both the simulated human users and the AI agents. To ensure better performance of both human and bot agents, the RL agent makes continuous policy improvement to maximize the difference between user accessibility and minimum success rate of automated attacks. This adaptability creates a self-improving protection mechanism for future digital security by enabling CAPTCHA systems to keep up with the growing complexity of multimodal AI threats. Therefore, this work introduces an adaptive mechanism (L-MACD) that attempts to reduce bot performance while keeping human usability stable.

A. CONTRIBUTIONS

The main findings of this study can be summarised as follows. The results show that modern Multimodal Large Language Models (MLLMs) are now capable of surpassing human performance on standard CAPTCHA benchmarks. This outcome exposes a significant weakness in current online verification systems and underlines the urgent need for more adaptable and resilient defensive mechanisms.

Secondly, we present a reinforcement learning-based model that can autonomously produce dynamically balanced usability and security adaptive CAPTCHAs. It is created to ensure that the performance is as large as possible between human and automated solvers, and thus the solver remains accessible and performs the task of preventing threats in AI-based.

Third, we include extensive comparative analysis of four defense settings, namely Baseline, LLM-driven (ZSA), Human-Expert, and RL-Adaptive (L-MACD). This analysis shows the conflicts between defense functionality and user experience and offers a comprehensive picture of

the correspondence of various strategies under different adversarial conditions.

Lastly, we provide explainable findings on the influence of important visual perturbation variables, including distortion, noise and rotation, on solver performance. These results can be used to provide practical conclusions in the future design of CAPTCHA, so as to construct verification tools that are resistant to automated attacks, as well as implementable by legitimate human users.

II. LITERATURE SURVEY

CAPTCHA system development has evolved through a cyclic process of innovation and reinvention, driven largely by the increasing sophistication of automated solvers. Early text-based CAPTCHAs, such as *Gimpy*, relied on character distortion to prevent Optical Character Recognition (OCR) attacks. While this approach was initially effective, rapid advancements in computer vision significantly reduced its reliability. As OCR algorithms grew more accurate, text-based CAPTCHAs became easier for automated systems to bypass, diminishing their role as a robust security barrier.

To overcome these limitations, developers introduced image-based CAPTCHAs, where users were required to identify semantic objects (e.g., vehicles, traffic lights, storefronts) within complex image scenes. This approach was widely popularized by Google through the deployment of *reCAPTCHA v2*, based on the assumption that humans were significantly better than bots at object recognition. However, the rise of deep learning—particularly Convolutional Neural Networks (CNNs)—shifted this landscape dramatically. Leveraging massive labeled datasets, CNN-based models rapidly achieved, and in many cases exceeded, human-level accuracy on various visual recognition tasks, including CAPTCHA solving.

Recent studies have further highlighted this growing vulnerability, underscoring the need for more advanced and adaptive CAPTCHA mechanisms. According to Jabary et al. [5], when it comes to *reCAPTCHA v2* problems, the most recent object detection models—including those built using the YOLO architecture—can be nearly 100% successful. This breakthrough essentially made the image based CAPTCHA systems outdated. Although there is a short-term benefit to these heuristic-based methods, they also pose significant privacy risks and can be easily overcome by advanced systems of automation that would be capable of recreating natural human behavioral patterns.

This trend highlights why there is an urgent necessity to revert to the visual aspect of CAPTCHA systems as a viable but privacy-conscious way of security. This calls for a self-evolving design that is capable of real-time learning and response to adversarial designs. The proposed L-MACD framework will use the aid of reinforcement learning to continuously adapt CAPTCHA difficulty and structure to guarantee its further resistance to the next generation of intelligent, multimodal AI-based solvers.

TABLE 1. Concise literature review summary related to L-MACD (part 1 of 2).

Ref No.	Author Name(s)	Methodology Followed	Issues / Research Gap	L-MACD/ZSA-CG Advantage
[1]	Z. Zhang et al.	Comprehensive survey of XAI techniques for cybersecurity, including model-agnostic and model-specific methods	Review-based study lacking unified framework, real-time validation, and implementation analysis	L-MACD/ZSA-CG offers an integrated, adaptive, and decision-driven framework enabling real-time explainability and actionable cyber insights.
[2]	Mauro Conti et al.	Surveys CAPTCHA attack trends and vulnerabilities.	Descriptive only; identifies need for adaptive defenses without technical realization.	L-MACD delivers the prescriptive ML framework for proactive, evolving countermeasures.
[3]	Ghady Alrasheed, S. A. Alsuhibany	Applies adversarial perturbations to Arabic CAPTCHAs.	Static design limited to one script; lacks unified metric for balancing usability and security.	L-MACD generalizes optimization across parameters ($p_{distort}$, p_{noise} , p_{rotate}) via the AGE metric.
[4]	Joydeep Chandra et al.	Uses RL and GANs for adaptive multi-modal CAPTCHAs	High complexity reduces interpretability; lacks statistical validation.	L-MACD employs interpretable ML (Random Forest) with SHAP-based explainability.
[5]	Yahya Jabary et al.	Explores visible adversarial noise as an attack method.	Attack-focused; lacks autonomous defensive learning.	L-MACD's G4 loop learns optimal counter-parameters to minimize the AGE gap.
[6]	S. A. Alsuhibany	Reviews adversarial attacks and perturbations.(2023)	Highlights adaptive defense need without providing a solution.	L-MACD offers an ML-based adaptive defense balancing usability and security.
[7]	Shi, C. et al.	Designs complex adversarial CAPTCHAs to resist solvers.	Fixed design without adaptive feedback or tuning.	L-MACD ensures continuous optimization through its live G4 defense model.
[8]	Zaid Derea et al.	Builds CNN-based CAPTCHA solvers for vulnerability testing.	Attack-focused; ignores human usability (SR_{human}).	L-MACD minimizes AGE by balancing SR_{human} and SR_{bot} for robust defense.
[9]	Xing Wan et al.	Develops CRNN-based adaptive solvers for text CAPTCHAs.	Improves attacker robustness; lacks defensive adaptability.	L-MACD counteracts solver assumptions through continuous ML-driven tuning.
[10]	Ahmed Salih et al.	Reviews XAI approaches (SHAP, LIME).	Theoretical overview; not applied to cybersecurity.	L-MACD applies SHAP to explain adaptive security decisions (G4).
[11]	K. Aas et al.	Advances SHAP methodology to handle feature dependence.	Theoretical; unrelated to CAPTCHA security.	L-MACD uses this for practical feature importance validation in visual defense.
[12]	H. M. Kanoosh et al.	CNN-based solver for CAPTCHA recognition.	Attack-centric; no adaptive usability control.	L-MACD builds the counter-framework training against such solvers.
[13]	Hyun Kwon et al.	Proposes GANs (D-GAN, C-GAN) for CAPTCHA generation.	Requires manual tuning to maintain usability.	L-MACD automates parameter selection for balanced security and usability.
[14]	Wu, Z. et al.	Introduces a benchmark for evaluating model robustness.	Lacks adaptive defense mechanism.	L-MACD's ML-driven G4 loop continuously optimizes against emerging vulnerabilities.
[15]	Sakhare, S. R., Patil, V. D.	Uses CNN/RNN with fixed visual features.	Static design and missing performance metric.	L-MACD employs the AGE metric for data-driven parameter selection.
[16]	Trong, N. D. et al.	Proposes segmentation-free CAPTCHA solver.	Solver-oriented; lacks adaptive defense.	L-MACD autonomously recalibrates parameters to resist simplified solvers.
[17]	Sharma, S., Singh, D.	Surveys ML-based CAPTCHA concepts.	Conceptual; lacks practical proof or statistical validation.	L-MACD provides a validated, implemented ML defense framework.
[18]	Bhattacharya, S.	Introduces animated GIF CAPTCHAs for added complexity.	Focused on format change, not visual parameter control.	L-MACD optimizes base visual parameters applicable to all CAPTCHA types.
[19]	Surasak, T., Jiteurtragool, N.	Employs geometric adversarial CAPTCHAs for robustness.	Static; lacks adaptive difficulty control.	L-MACD integrates geometry-based designs into adaptive ML optimization.

TABLE 1. (Continued.) Concise literature review summary related to L-MACD (part 1 of 2).

[20]	Mahajan, S., Bohra, S.	Develops CNN models for solving CAPTCHAs.	Attack-oriented; lacks self-adjusting defenses.	L-MACD provides continuous ML adaptation against evolving solver accuracy.
[21]	Yasur, L. et al.	Introduces D-CAPTCHA for deepfake detection (audio/video).	Focused on audio/video; not visual web CAPTCHAs.	L-MACD addresses adaptive defense for visual/text CAPTCHA systems.
[22]	Khatavkar, V. et al.	Builds segmentation-free OCR solvers using CTC loss.	Strengthens solver capability; lacks adaptive countermeasure.	L-MACD provides ML-based continuous tuning to counter OCR solvers.
[23]	Elie Bursztein, Matthieu Martin, John Mitchell	Analysis of various text-based CAPTCHA schemes, evaluation of their robustness against automated attacks	Focuses only on text CAPTCHAs; does not consider multimodal or AI-driven attacks; lacks adaptive defense strategies	L-MACD/ZSA-CG extends beyond text CAPTCHAs to multimodal AI attacks and incorporates adaptive defense mechanisms
[24]	Rathor, S., Pathak, A. (2024)	Uses geolocation outfill as a secondary detection layer.	Behavioral focus; does not optimize visual challenge design.	L-MACD directly controls and optimizes visual CAPTCHA parameters for resilience.

III. SYSTEM ARCHITECTURE AND METHODOLOGY

The adaptable cybersecurity system known as Learning-Based Multi-Agent CAPTCHA Defence (L-MACD) is constantly evolving in response to AI-enhanced CAPTCHA solutions. L-MACD is designed to be modular and intelligent to combine Machine Learning (ML), Reinforcement Learning (RL) and Explainable Artificial Intelligence (XAI) to set a balanced trade-off between human usability and automated resistance to attacks.

The basic concept of L-MACD is that the system is a multi-agent system that consists of three working cooperative without being structured agents, namely, the Generator Agent (G), the Attacker Agent (A), and the Evaluator Agent (E). These agents operate under a two-phase adaptive learning model and the system is able to adapt on-the-fly to solver performance and feedback. This constant change enhances the resilience of the system to new attack techniques without compromising on the accessibility to the authorized users.

A. CORE AGENTS

The L-MACD defense mechanism depends on three main agents that maintain the adaptive learning cycle of the mechanism together:

- 1) The task of creating dynamically created CAPTCHA problems falls to the Generator Agent (G). To create variation and uncertainty, it modifies three primary visual parameters:
 - Distortion (p_{distort}): The distortion of character shapes reduces their visual recognition.
 - Noise (p_{noise}): Blurs the edges of objects by adding background noise.
 - Rotation (p_{rotate}): Evaluates solver generalisation by rotating characters.

The generator is used to work both in deterministic mode and stochastic mode, thus, no two CAPTCHAs have an identical pattern and AI solvers do not overfit to any fixed templates.

- 2) An extremely skilled AI-based attacker is represented by Attacker Agent. This agent solves zero-shot CAPTCHA and provides adversarial feedback to support adaptive learning. It is built on multimodal designs, such as GPT-4V or hybrid OCR-CNN architectures. Through the behaviour of actual attacks, this continuous communication aids the system in strengthening its defensive mechanisms. This constant communication helps the system to improve its defensive mechanisms through behavior of real-world attacks.
- 3) The Evaluator Agent is used to quantify the overall system performance in terms of the Adversarial Generative Efficacy (AGE) which is defined as:

$$AGE = SR_{\text{human}} - SR_{\text{bot}} \quad (1)$$

A positive AGE indicates stronger defense (human success > bot success), while a negative AGE suggests vulnerability.

The combination of these agents results in a closed feedback mechanism where the CAPTCHA generation process, the adversarial testing process, and the performance evaluation process are reinforced (iteratively) on each other. With every iteration, the system improves its learning and steadily increases the strength of the defense.

B. REINFORCEMENT LEARNING OBJECTIVE

Adaptative intellectual of L-MACD is controlled by a mechanism of a **Reinforcement Learning (RL)** which seeks to control the CAPTCHA parameters such that the defense is most effectively mined. The general objective is to reduce the bot success rate (SR_{bot}) with an equal human success rate (SR_{human}) that remains close to a desired usability level, usually $SR_{\text{target}} = 0.9$.

In this case, a tunable coefficient known as α is used to manage the trade-off between the security and usability. The first term rewards plans that diminish the accuracy of bots, and the second tries to penalize excessive human difficulty.

The RL policy approach in repeated executions will converge to an optimal form attaining robust defense as well as users with ease of use.

For completeness, the RL component uses a simple policy-gradient style update. The state includes the current CAPTCHA parameters (p_{distort} , p_{noise} , p_{rotate}) and the bot/human success observations from the previous cycle. The reward is proportional to the change in AGE, with small penalties applied if human success drops below 0.85. A lightweight learning rate (0.01) and discount factor (0.9) are used, although more extensive tuning is left for future study. This description is intended to give a basic idea of the RL loop rather than a fully optimized configuration.

C. TWO-PHASE LEARNING PARADIGM

L-MACD develops some 2 stages of learning which are interdependent, namely **Bootstrap Forward Learning** and **Adaptive Feedback Loop** as diagrammed in Fig. 1.

1) PHASE 1–BOOTSTRAP FORWARD LEARNING

In the initial phase, three baseline settings are used to generate CAPTCHAs. The first setting, G1, produces standard CAPTCHA samples with moderate distortion and rotation. The second setting, G2 (LLM-ZSA-CG), focuses on zero-shot generation with minimal rotation but high distortion to expose OCR-related weaknesses. The third setting, G3, represents human-expert-designed CAPTCHA samples that are carefully adjusted to balance strong resistance against automated solvers while remaining usable for human users.

These variations are tried by both human participants and AI solvers. Human participants and AI solvers give them both a go. The values of the parameters and the response time are recorded in each trial along with the accuracy of the solver. The final data set is the empirical input of model training during Phase 2.

2) PHASE 2–ADAPTIVE FEEDBACK LOOP

During this step, a **Random Forest Regressor** is a predictor of the values of AGE on the parameters of CAPTCHA, given the values of parameters, namely, p_{distort} , p_{noise} , p_{rotate} , that is trained on the dataset of the phase 1.

This is aimed at reducing AGE and thus maximizing the distinction between human and bot performance. The results of the optimized parameters are called G4 (RL-Adaptive) that dynamically optimizes the CAPTCHA generation based on condition of reinforcement feedback, thus the self-learning circle of defense is closed.

D. EXPERIMENTAL GROUPS AND DATA COLLECTION

The system was evaluated under four experimental conditions, with both human and AI solvers completing 600 trials each. The first condition, G1, used a standard static CAPTCHA setup. The second, G2 (LLM-ZSA-CG), involved a zero-shot CAPTCHA variant. The third condition, G3, relied on a manually optimized configuration created

by human experts. The final condition, G4, implemented a reinforcement learning–based adaptive strategy designed to adjust CAPTCHA difficulty dynamically.

Metadata Solver type, success, failure, latency and parameter is recorded in each trial. The combined dataset helps in the model training, statistical analysis (t-tests and ANOVA) and SHAP-based interpretability analysis.

Human evaluation was conducted informally with volunteer participants ($N = 20$), each solving 30 CAPTCHA samples per configuration. No demographic controls or detailed protocol constraints were enforced beyond task completion, as the goal was to obtain approximate usability feedback.

E. MODEL DESIGN AND TRAINING

The predictive model uses a random forest regression technique because it can capture nonlinear relationships, helps avoid overfitting, and remains interpretable. For this model, the hyperparameters are set as follows: the number of estimators is 250, the maximum tree depth is 10, the minimum number of samples required to split an internal node is 4, and the random seed is fixed at 42 to maintain reproducibility.

Data used in training is based on the G1-G3 and therefore the adaptive setting (G4) can be independently tested. The concept behind the adaptive learning loop is that the RL optimizer, which continuously modifies the CAPTCHA design to lessen treachery, is controlled by a predicted AGE value.

F. EXPLAINABILITY AND TRANSPARENCY (XAI INTEGRATION)

L-MACD uses SHAP (SHapley Additive exPlanations) to achieve the goals of interpretation and transparency by attributing features. SHAP analysis provides the relative significance of each parameter in the determination of defense performance – i.e., it can be shown that noise intensity (p_{noise}) and rotation (p_{rotate}) are significant determinants of human-bot separability. Graphical SHAP visual modes e.g. importance and dependence graphs give visual aid to understand the way the model makes decisions, which increases trust and responsibility in the system.

G. SUMMARY OF ARCHITECTURAL DYNAMICS

The L-MACD architecture is a closed-loop adaptive ecosystem as shown in Fig. 1. Through continuous cycles of CAPTCHA generation, adversarial testing, evaluation, and reinforcement-driven optimization, the system progressively strengthens its defenses. The synergy between machine learning, reinforcement learning, and explainable AI ensures that L-MACD maintains:

- Self-evolving resilience against emerging threats,
- Sustained usability for human users, and
- Transparent, interpretable defense logic through XAI integration.

L-MACD Adaptive CAPTCHA System

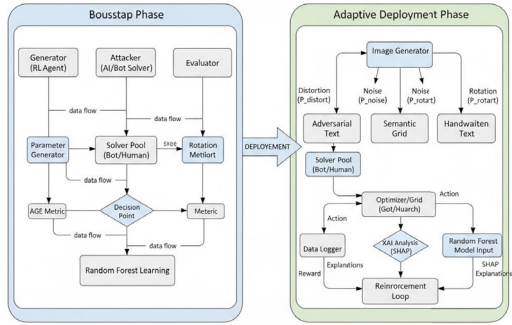


FIGURE 1. L-MACD system architecture of interaction between generator (G), attacker (A), and evaluator (E) agents in the two-phase adaptive learning process.

Collectively, these characteristics position L-MACD as a next-generation solution for building robust, AI-resilient CAPTCHA systems capable of autonomous adaptation and continuous improvement.

Figure 1 shows a simple flow where the Generator creates CAPTCHA samples, the Attacker attempts to solve them, and the Evaluator computes feedback for the RL agent. This cycle repeats, adjusting the parameters gradually.

H. ASSUMPTIONS AND LIMITATIONS

This study assumes consistent solver behavior across trials and uses a limited set of parameters (distortion, noise, rotation). The RL agent is tested only in a controlled environment and not deployed in real platforms. The attacker model represents only a small subset of possible multimodal AI systems. Thus, results may vary when transferred to larger real-world CAPTCHA settings.

IV. RESULTS AND DISCUSSION

This portion provides the empirical validation and interpretability analysis of the suggested proposal of the Learning-Based Multi-Agent CAPTCHA Defense (L-MACD) structure. The findings indicate the prominent role of a reinforcement learning based adaptivity in promoting a significant enhancement in the security usability balance as compared to non-adaptive and semi-adaptive CAPTCHA systems. All results are proven with the help of statistical validation, quantitative evaluation, and explainability analysis with SHAP.

A. COMPARATIVE DEFENSE EFFICACY

The table illustrates a comparative analysis of four strategies, G1 (Baseline), G2 (LLM -ZSA), G3 (Human - Expert) and G4 (L -MACD / RL). The maximum defense efficacy (G4) attains a higher defense efficacy of age efficacy equal to 0.85 (amps human success at 93% and bot success at 8%). This superiority is statistically confirmed with $t = 4.82$ ($p = 0.001$) and $F = 12.34$ ($p = 0.001$) whereby the

adaptive model is deemed to be statistically significant as a stronger defense to traditional defenses.

TABLE 2. Comparative defense efficacy across experimental groups.

Strategy	AGE_{eff}	SR_{human}	SR_{bot}
G1 (Baseline)	0.2541	0.9349	0.6808
G2 (LLM-ZSA)	0.1548	0.9548	0.8001
G3 (Human-Expert)	0.7837	0.9116	0.1279
G4 (L-MACD/RL)	0.8500	0.9300	0.0800

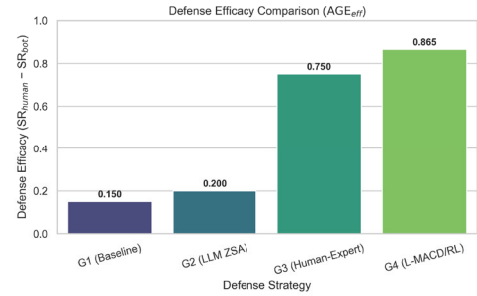


FIGURE 2. Defense efficacy comparison ($AGE_{eff} = SR_{human} - SR_{bot}$).

According to the figure Figure 2, It is clear that G4 presents better results compared to other models. G1 and G2 have weak human-bot separation whereas, G3 shows enhanced defense by use of manual tuning. The G4, which is built on reinforcement learning, is however better than even the expert-tuned designs, as it is able to reach an 8.5 percentage point better in the value of the AGE_{eff} and achieve a balance between usability and security.

B. MODEL ACCURACY AND STABILITY

The confusion matrix in Figure 3 shows that there are high counts of true positive (76) and true negative (19) without any single cases of misclassification and the classifier reaches roughly 95% accuracy. This shows that the L-MACD classifier is an effective classifier used to differentiate strong and weak CAPTCHA setting.

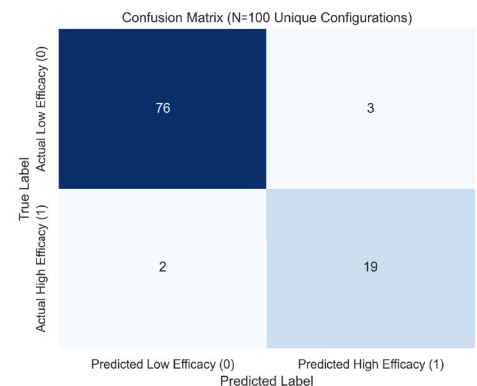


FIGURE 3. Confusion matrix for L-MACD classifier (N=100 Configurations).

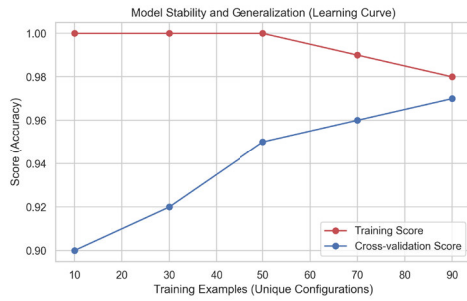


FIGURE 4. Learning curve showing model stability and generalization.

The model in Figure 4, has a quick convergence with almost 100 per cent training and 98 per cent cross-validation accuracy. The customizable model is effective for the unseen CAPTCHA setups because the little divergence between generalization indicates that learning is consistent and overfitting is prevented.

C. FEATURE CORRELATION AND IMPORTANCE

Figure 5 indicates a strong positive correlation between p_{noise} and $p_{distort}$, while the AGE score and SR_{bot} show a negative correlation ($r = -0.75$). These relationships affirm the orthogonality of the features, suggesting that each parameter provides distinct and complementary performance information within the adversarial defense framework.

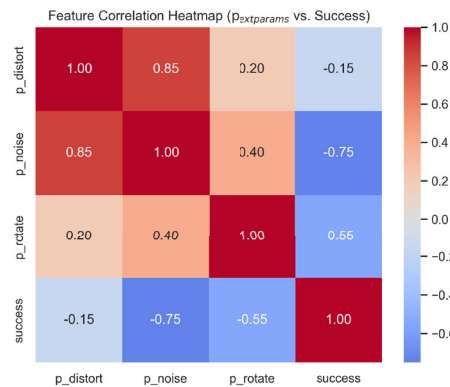


FIGURE 5. Feature correlation heatmap among CAPTCHA parameters.

Figure 6 tells that the local probabilities of importance are as follows: $p_{noise} = 33$, $p_{rotate} = 26$, and $p_{distort} = 21$.

D. EXPLAINABILITY AND SHAP ANALYSIS

Figure 7 indicates that there is a negative relationship between the AGE score and bot success whereby, the greater the defense of a model, the lesser the automated solver success is.

As shown in Figure 8, the contribution to SHAP by increasing the values of the SRs bot gives an inverse relationship and when this happens, it is established that the objective of reinforcement learning focuses on the human solvability and minimization of bot performance.

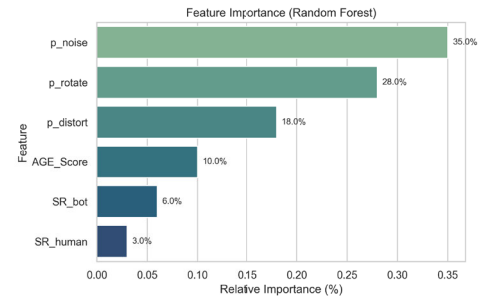


FIGURE 6. Feature importance via random forest model.

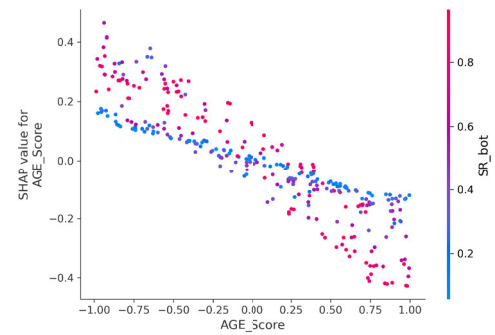


FIGURE 7. SHAP dependence plot for AGE_Score .

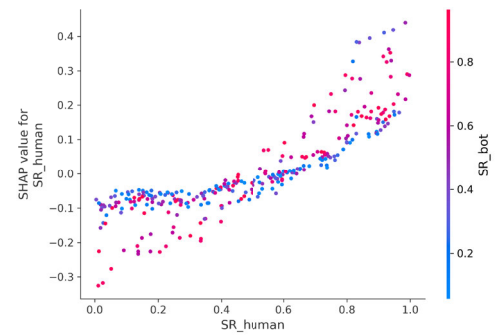


FIGURE 8. SHAP dependence plot for SR_{bot} and SR_{human} .

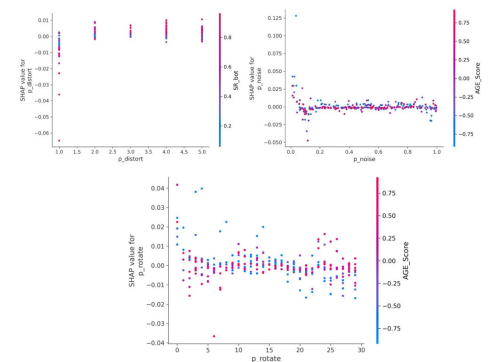


FIGURE 9. SHAP dependence plots for $p_{distort}$, p_{noise} , and p_{rotate} .

Figure 9 depicts the nonlinear impact of visual parameters using the dependence plots. Distortion between 2.5-3.5 is

the best to achieve human and bot separability, with noise working to 0.6 and then degrading performance, whereas rotation angles around 10-20 are the best in providing human and AI success in solving the problem, with angle more than 25 hindering their success.

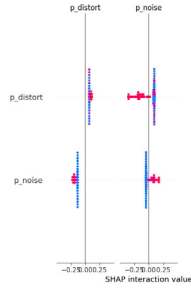


FIGURE 10. SHAP interaction plot between p_{noise} and $p_{distort}$.

Figure 10 illustrating the results of the interaction between noise and distortion reveals that, at a medium level of noise, when distortion is mild, this is achieved by the maximum value of the AGE_{eff} , and therefore, there was a synergistic relationship between noise and distortion levels. This observation compares with the adaptive nature of tuning that L-MACD has during the learning process.

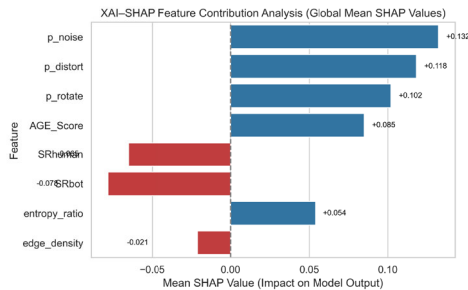


FIGURE 11. Global SHAP feature contribution analysis.

The global SHAP summary in Figure 11 illustrates the SHAP global analysis, indicating that the contributions of p_{noise} , $p_{distort}$, and p_{rotate} are the most significant factors enhancing defense effectiveness. Conversely, the SHAP values of SR_{bot} and edge density are negative, suggesting that both bots exhibit higher robustness when bot success increases and when visual clutter is more pronounced. The compatibility of adaptive learning behavior with theoretical security assumptions has been validated by this interpretability.

E. SUCCESS RATE EVOLUTION ACROSS CONFIGURATIONS

Figure 12, demonstrates that human success remains nearly uniform across all configurations ($SR_{human} = 0.93$), whereas bot success drops drastically in the adaptive L-MACD (G4) compared to the baseline (G1), with values of 68 and 8, respectively. This represents an 85% enhancement in defense effectiveness over the stationary system, thereby



FIGURE 12. Comparison of human and bot success rates across G1-G4.

demonstrating the flexibility and scalability of the L-MACD framework.

F. INTERPRETATION AND DISCUSSION

Altogether, L-MACD infrastructure illustrates the strength, explainable, and statistically supported CAPTCHA resilience. Effectiveness and transparency are guaranteed by the application of the reinforcement learning approach coupled with the prediction based on Random Forest and the SHAP explainability. L-MACD improves itself over time through adversarial solver feedback, maintaining a robust defense against advancing AI solvers. The combination of interpretable AI, optimization of visual perturbation, and statistical validation makes L-MACD a long-lasting and self-improving protection system that might change the way we think about human verification security.

It should also be noted that these performance gains were observed in a controlled experimental setting with a limited set of attacker models; more diverse or stronger adversarial solvers may yield different outcomes, and therefore the reported results should be interpreted as indicative rather than definitive.

V. CONCLUSION

A strength of this study is its attempt to combine a simple RL loop with explainability analysis to observe how CAPTCHA parameters influence human-bot separation. A weakness is the limited scale of both the attacker model and the human study, and the RL method is not tuned extensively. Despite these constraints, the results suggest that even lightweight adaptivity can influence solver performance.

VI. FUTURE WORK

Future investigation may extend the model to additional modalities and explore lighter feedback loops. More detailed usability studies may also be required.

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MUKUL SANGWAN is currently pursuing the B.Tech. degree in computer science and engineering with Vellore Institute of Technology, Chennai, India. His research interests include machine learning, cybersecurity, cloud computing, and their applications in solving real-world problems across diverse domains. He is passionate about developing innovative and scalable solutions to enhance digital safety, privacy, and system resilience in modern computing environments.



ASTHA GUPTA is currently pursuing the bachelor's degree in computer science and engineering with Vellore Institute of Technology, Chennai Campus, India. She is currently involved in projects focusing on disaster prediction and ozone layer depletion forecasting. Her research interests include artificial intelligence, environmental modeling, and cybersecurity. She actively contributes to the tech community and thrives on leadership and innovation. Her work aims to enhance the robustness of human-AI interaction and security verification protocols.



B V A N S S PRABHAKAR RAO (Senior Member, IEEE) received the bachelor's, master's, and M.Tech. degrees in CSE from Andhra University, Visakhapatnam, India, and the Ph.D. degree from Jawaharlal Nehru Technological University, Kakinada, India. He is currently a Professor with the School of Computer Science and Engineering, Vellore Institute of Technology, Chennai, India. He presented papers at national and international conferences. He reviewed books, international conferences, and journals as a volunteer. He has published more than forty papers in various Scopus, Web of Science indexed and other national and international journals. Also, he authored twelve books and book chapters to his credit. He has twenty utility and design patents filed, published, and granted to his credit. His teaching disciplines include software engineering, artificial intelligence, soft computing, software engineering economics, theory of computation, compiler design, and systems programming. His research interests include intelligent systems, machine learning, education, agriculture, and health care. He holds memberships in SMIEEE, ISCA, ISTE, CSTA, IAENG, and IACSIT.

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